



FROM ZERO TO PRO - MODULE 03

Heat & Cooking Methods

Every dish is an application of heat; learn the methods and you can cook almost anything without a recipe.

Weeks 5-6 - watch, study, cook, reflect

By the end of this module

- 1 Distinguish dry-heat (saute, sear, roast, grill, fry) from moist-heat (steam, poach, simmer, braise) methods.
- 2 Control a pan: preheating, fond, deglazing, and the Maillard reaction.
- 3 Choose the right method for a given ingredient.
- 4 Use a thermometer and your senses to judge doneness.

LESSON 3.1

How Heat Moves and Why Food Browns

Objective: Understand conduction, convection, radiation, and the browning reactions.

Key points

- Conduction = contact, convection = moving fluid, radiation = waves.
- Maillard browning (proteins + sugars) builds savory crust above ~300 deg F.
- Caramelization is sugars browning alone - sweet and nutty.
- Wet food steams and will not brown until the surface dries.

Watch out for

- Crowding a pan so food steams in its own moisture and never browns.
- Adding salt to meat right before searing, drawing moisture to the surface.
- Expecting color from a method that stays at or below boiling.

LESSON 3.2

Dry-Heat Methods

Objective: Control sauteing, searing, roasting, grilling, and frying.

Key points

- Preheat the pan before adding oil and food; a cold pan means no sear.
- Let seared food sit undisturbed - it releases from the pan when crusted.
- Do not crowd; cook in batches so heat stays high and food browns.
- Fry at 325-375 deg F; too-cool oil makes greasy food.

Technique

- 1 Preheat**
Heat the empty pan, then add a thin film of oil until it shimmers.
- 2 Lay away**
Place food down away from you; do not move it.
- 3 Wait**
Let the crust form; the food will release when ready.
- 4 Flip once**
Turn a single time for an even, deep sear.

LESSON 3.3

Moist-Heat and Combination Methods

Objective: Steam, poach, simmer, and braise with control.

Key points

- Poach at 160-180 deg F (a bare shiver), not a rolling boil.
- Simmer = gentle bubbles (~185-205 deg F); boil = rolling (212 deg F).
- Braise = sear, then slow-cook partly submerged in liquid.
- Low and slow turns tough collagen into tender gelatin.

Technique

- 1 Sear**
Brown the meat well on all sides for flavor and color.
- 2 Build**
Soften aromatics, deglaze, add just enough liquid to come partway up.
- 3 Cover**
Lid on, low oven or low burner, until fork-tender.
- 4 Reduce**
Strain and reduce the liquid into a glossy sauce.

LESSON 3.4

Doneness, Resting, and the Pan Sauce

Objective: Judge doneness, use carryover and resting, and build a sauce from fond.

Key points

- Use a thermometer; pull a few degrees early for carryover.
- Rest meat so juices redistribute - slicing too soon loses moisture.
- Fond (browned bits) is the base of a great pan sauce.
- Deglaze, reduce, then finish with butter for body and shine.

Technique

- 1 Rest**
Move the protein to a board; tent loosely while you make the sauce.
- 2 Pour off**
Drain excess fat, leaving the fond and a little fat behind.
- 3 Deglaze**
Add liquid and scrape up the fond as it dissolves.
- 4 Reduce and finish**
Simmer to concentrate, then swirl in cold butter off heat.

Practice & homework

1

Cook the same vegetable three ways (roast, saute, steam); taste side by side and journal.

2

Build and reduce a simple pan sauce from fond after searing.

3

Sear a protein to a target internal temperature using a thermometer.

Watch & learn (free)



Ethan Chlebowski (the 'why' behind cooking)

<https://www.youtube.com/@EthanChlebowski>



J. Kenji Lopez-Alt (food science & technique)

<https://www.youtube.com/@JKenjiLopezAlt>



America's Test Kitchen (tested technique)

<https://www.youtube.com/@AmericasTestKitchen>



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Next up - Module 04: Stocks, Soups & the Five Mother Sauces

FROM ZERO TO PRO - THE SELF-TAUGHT CHEF PROGRAM